

AHAN MONDAL

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EDUCATION

Kalinga Institute of Industrial Technology

Bachelor of Technology in Computer Science — CGPA: 8.83 (5 Semesters)

Bhubaneswar, Odisha

Aug. 2023 – Present

Salt Lake School

Class 12 (ISC) – 89.5%(Best of 4) — Class 10 (ICSE) – 95.6%(Best of 5)

Kolkata, West Bengal

2008 – 2023

SKILLS

Languages: Java, Python, C, SQL, Shell Script

Core Concepts: DSA, OOP, DBMS, CN, Deep Learning, Graph Neural Networks, LangGraph

Frameworks & Libraries: PyTorch, PyTorch Geometric, Pandas, NumPy, Scikit-learn, CatBoost, Matplotlib, Streamlit

Databases: SQLite, SQL(Oracle)

Tools: Power BI, Excel, Git, Linux, Vercel, Vite, Vim, VS Code, Jupyter Notebook

PROJECTS

MedPath AI [Live] | *FastAPI, React 19, LangGraph, Gemini 2.5 Flash, Supabase, JWT*

May 2026

- Led end-to-end chatbot development for a 3-person team, designing the 4-node LangGraph state graph (intent → provider → cost → response), selecting Gemini 2.5 Flash for clinical intake, and delivering emergency bypass detection across 500+ hospital records.
- Partnered with backend engineer on a multi-factor hospital scoring engine using Pandas and Haversine distance, with comorbidity-based cost multipliers for diabetes (+12%) and cardiac history (+18%); enabling Supabase-backed loan application and status tracking with 3-second polling updates.
- Coordinated with backend engineer to connect a loan pre-qualification pipeline with 4 underwriting checks and PDF data extraction, reducing manual eligibility review to under 1 minute with instant GREEN/YELLOW/RED decisions

Vehicle Insurance Fraud Detection | *Python, PyTorch, CatBoost, PyTorch Geometric, Scikit-learn*

Mar. 2026

- Architected a multi-modal fusion model as part of a 4-person team, combining 16-dimensional CatBoost tabular embeddings, 16-dimensional LSTM temporal embeddings, and 32-dimensional GCN graph embeddings into a 128-64-1 DNN classifier.
- Owned data preprocessing and LSTM temporal branch in a collaborative pipeline; engineered stratified splitting, weighted loss functions, and focal-loss experimentation achieving ROC-AUC 0.8291 and PR-AUC 0.2259.
- Constructed a threshold-tuning workflow over 91 candidates from 0.05 to 0.95, selecting a precision-first threshold of 0.54 at 0.80 recall and a cost-optimal threshold of 0.39 with FP/FN cost 1:25 and business cost 1489.0.

Classroom Scheduling System | *Python, SQLite, pytest, OOP*

Jan. 2026 – Feb. 2026

- Developed a scheduling platform with 4 core entities (rooms, teachers, courses, timeslots) and 1 schedule-mapping layer to automate timetable generation and classroom allocation.
- Implemented scheduling logic with 5 deterministic conflict checks (room overlap, teacher overlap, duplicate course-slot, capacity mismatch, projector mismatch) plus a best-fit auto-assignment routine that searches valid timeslot-room combinations.
- Modelled a 5-table normalized SQLite schema with 3 foreign-key relationships into schedules, 3 composite UNIQUE constraints on slot collisions, and transactional writes via context-managed DB connections.
- Engineered and validated a Pytest suite with 13 automated test cases (13/13 passing, 100% pass rate, 1.88s runtime) covering manual scheduling, auto-assignment, and conflict-reporting behaviors.

ACHIEVEMENTS

AlgoUniversity Graphs Contest — Rank 10

- Secured Rank 10 within the college in the Graphs Contest organized by AlgoUniversity.
- Qualified for accelerator camp benefits and scholarship consideration based on contest performance.